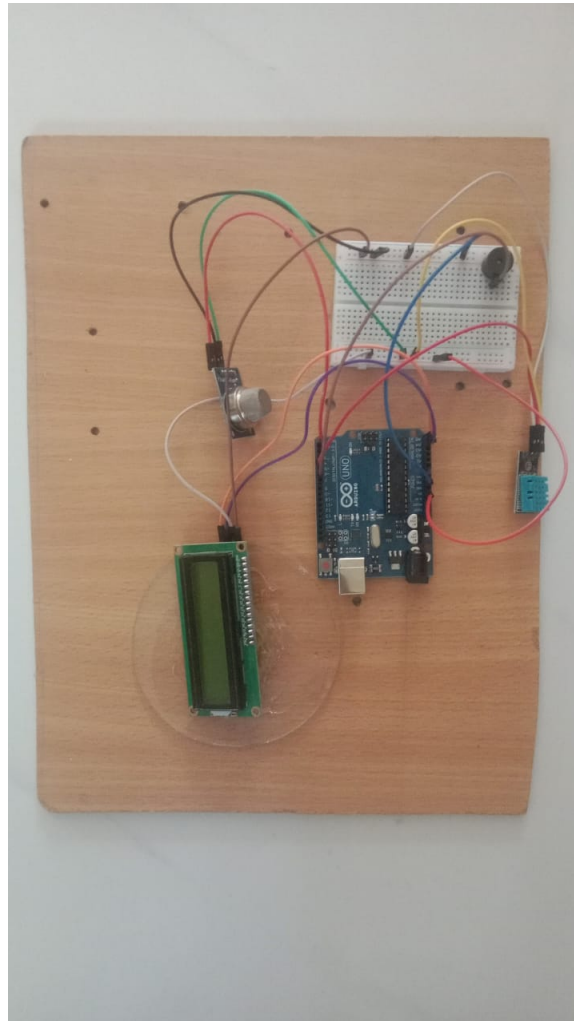


# FIRE AND SMOKE DETECTION SYSTEM WITH ALARM USING ARDUINO

Arduino-Based Safety Monitoring Project



**Prepared by**

Atchaya G • Sowmiya S • Lalitha Sree P • Janani N

**Department of Electrical and Electronics Engineering**

VSB College of Engineering Technical Campus, Coimbatore

Project Report

# 1. Introduction

Fire accidents pose serious risks to life and property, making early detection an important safety requirement. This project presents an Arduino-based fire and smoke detection system designed to monitor environmental conditions and provide an immediate warning when unsafe conditions are detected.

The supplied project material describes the use of an Arduino controller, MQ-2 smoke sensing, flame sensing, a buzzer alarm and an LCD display. The accompanying prototype code also monitors temperature and humidity using a DHT11 sensor and compares temperature and gas readings with predefined limits.

## 2. Objectives

- Detect dangerous smoke/gas or fire-related conditions at an early stage.
- Provide an immediate audible warning using a buzzer.
- Display monitoring and alert information on an LCD.
- Develop a low-cost and practical safety monitoring system.
- Provide a platform that can be extended with wireless or IoT features.

## 3. Key Features

| Feature               | Description                                                   |
|-----------------------|---------------------------------------------------------------|
| Continuous monitoring | Sensors are read repeatedly by the Arduino.                   |
| Threshold-based alert | Unsafe readings trigger the alarm condition.                  |
| Audio indication      | Buzzer provides an immediate warning.                         |
| Visual indication     | LCD displays system status and alert information.             |
| Automatic reset       | Outputs return to normal when the measured condition is safe. |

## 4. Components Used

| Component                 | Purpose                                                       |
|---------------------------|---------------------------------------------------------------|
| Arduino Uno               | Main controller for reading sensors and controlling outputs.  |
| MQ-2 Smoke/Gas Sensor     | Monitors smoke/gas level in the environment.                  |
| IR Flame Sensor           | Used in the described project setup for flame detection.      |
| DHT11 Sensor              | The supplied Arduino code measures temperature and humidity.  |
| Buzzer                    | Provides audible alarm indication during an alert.            |
| 16×2 I2C LCD              | Displays temperature, humidity, gas reading and alert status. |
| LED                       | Provides an additional visual alert in the supplied code.     |
| Breadboard & Jumper Wires | Used to interconnect the prototype components.                |

## 5. System Architecture

The system follows a simple sensing–processing–alert architecture. Environmental information is collected by the sensing elements, processed by the Arduino Uno, and communicated to the user through the LCD, buzzer and LED.

| INPUT / SENSING                                                   | PROCESSING                                            | OUTPUT / ALERT                |
|-------------------------------------------------------------------|-------------------------------------------------------|-------------------------------|
| MQ-2 smoke/gas<br>IR flame sensor<br>DHT11 temperature & humidity | Arduino Uno<br>Threshold comparison<br>Decision logic | 16×2 I2C LCD<br>Buzzer<br>LED |

## 6. Working Principle

**Step 1:** Power is supplied to the prototype and the Arduino initializes the connected sensors, LCD and alert devices.

**Step 2:** The MQ-2 sensor monitors gas/smoke level. The described project setup also uses an IR flame sensor for flame detection.

**Step 3:** In the supplied Arduino program, the DHT11 sensor provides temperature and humidity values, while the gas sensor provides an analog reading.

**Step 4:** The Arduino continuously compares the measured temperature and gas value against predefined safety limits.

**Step 5:** If temperature exceeds 30 °C or the gas reading exceeds 400, the buzzer and LED are switched ON and the LCD displays an ALERT message.

**Step 6:** If the readings remain within the programmed limits, the buzzer and LED remain OFF and the LCD shows the measured values.

**Step 7:** The program repeats this monitoring cycle continuously with a one-second delay.

## 7. Alert Logic

**Alert condition:** Temperature > 30 °C OR Gas value > 400

**When alert occurs:** Buzzer = ON, LED = ON, LCD = ALERT

**Normal condition:** Buzzer = OFF, LED = OFF, LCD displays temperature, humidity and gas value.

## 8. Block Diagram

### Block Diagram

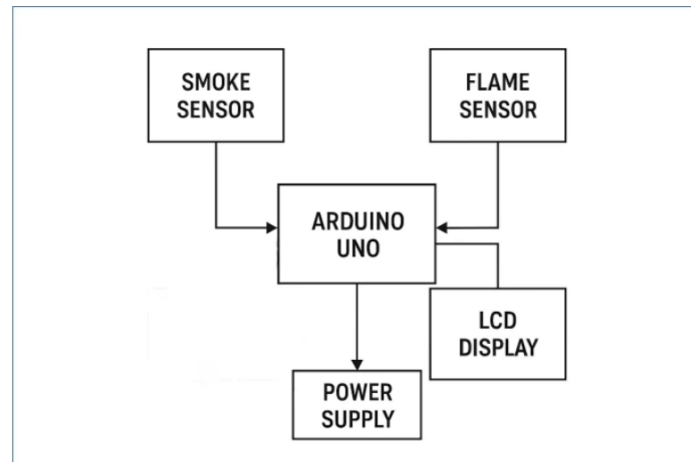


Figure 1. Block diagram from the supplied project material.

## 9. Circuit Diagram

### Circuit Diagram

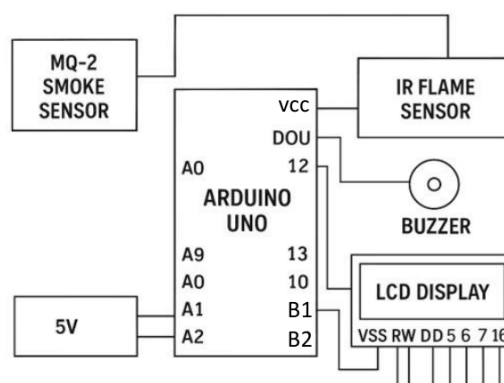


Figure 2. Circuit diagram from the supplied project material.

## 10. Prototype and Results

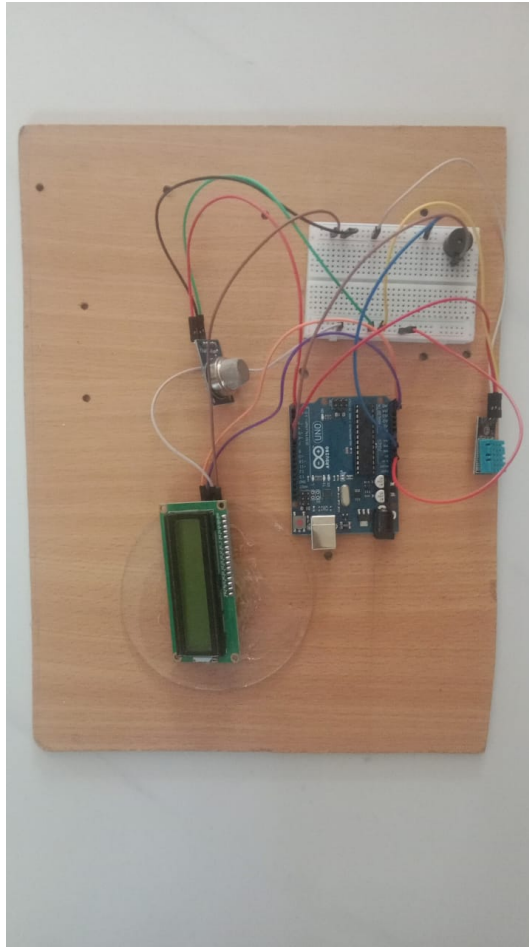


Figure 3. Working project prototype.

### Observed project results:

- Rapid detection of unsafe fire/smoke-related conditions.
- Stable and consistent system response.
- Audible and visual warning through the buzzer, LED and LCD.

## 11. Applications

- Residential and commercial safety systems.
- Industrial fire monitoring.
- Smart home and IoT safety networks.

## 12. Future Scope

The supplied project material identifies the following possible enhancements:

- GSM or Wi-Fi module for mobile notifications.
- Integration with AI-based camera detection.
- Solar-powered operation for remote zones.

Additional development could build on these ideas for remote monitoring and connected safety applications.

## 13. Conclusion

The Arduino-based Fire and Smoke Detection System provides a practical, cost-effective approach to early warning. By combining environmental sensing, threshold-based decision logic, LCD information and alarm outputs, the prototype can notify users when monitored conditions become unsafe. The project also provides a suitable foundation for future wireless, IoT and intelligent safety enhancements.

## 14. Project Team

Atchaya G

Sowmiya S

Lalitha Sree P

Janani N

Electrical and Electronics Engineering — VSB College of Engineering Technical Campus, Coimbatore

# 15. Arduino Program

The following program is reproduced from the Arduino code supplied with the project. It uses DHT11, MQ-2, buzzer, LED and I2C LCD interfaces.

```
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// ----- DHT11 Setup -----
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// ----- Gas Sensor & Alerts -----
#define GAS_SENSOR A0
#define BUZZER 5
#define LED 6

float tempLimit = 30; // Temperature threshold (°C)
int gasLimit = 400; // Gas threshold (analog value)

// ----- LCD Setup -----
LiquidCrystal_I2C lcd(0x27, 16, 2); // Change to 0x3F if needed (check via I2C Scanner)

void setup() {
  Serial.begin(9600);
  dht.begin();

  pinMode(BUZZER, OUTPUT);
  pinMode(LED, OUTPUT);
  digitalWrite(BUZZER, LOW);
  digitalWrite(LED, LOW);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Temp&Gas Detector");
  lcd.setCursor(0, 1);
  lcd.print("Initializing...");
  delay(2000);
  lcd.clear();
}

void loop() {
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();
  int gasValue = analogRead(GAS_SENSOR);

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C | Humidity: ");
  Serial.print(humidity);
  Serial.print(" % | Gas: ");
  Serial.println(gasValue);

  // ----- ALERT CONDITION -----
  if (temperature > tempLimit || gasValue > gasLimit) {
    digitalWrite(BUZZER, HIGH);
    digitalWrite(LED, HIGH);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ALERT");
    lcd.setCursor(0, 1);
    lcd.print("Temp:");
    lcd.print(temperature, 1);
    lcd.print((char)223);
    lcd.print("C");
  }
  else {
    digitalWrite(BUZZER, LOW);
    digitalWrite(LED, LOW);

    lcd.setCursor(0, 0);
    lcd.print("T:");
    lcd.print(temperature, 1);
    lcd.print((char)223);
    lcd.print("C H:");
    lcd.print(humidity, 0);
    lcd.print("% ");

    lcd.setCursor(0, 1);
    lcd.print("Gas:");
  }
}
```



```
    lcd.print(gasValue);  
    lcd.print("    ");  
}  
  
delay(1000);  
}
```